

Neuro-Sym Supporter: A Thoughtful Emotion Support Agent Integrating Neural and Symbolic Policy Learning

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Abstract

LLM-based empathetic dialogue systems enhance agents' emotional support capabilities. Previous approaches primarily relied on Chain-of-Thought (CoT) prompting to extract key dialogue cues and further strengthened the agent's sensitivity to these signals through supervised fine-tuning. However, such methods overly depend on the information extraction capability of LLMs, leading to unstable reasoning and limited interpretability. To simultaneously improve an agent's ability to proactively explore solutions through rational reasoning while attending to users' sensitive emotions via empathetic understanding, we propose **Neuro-Sym Supporter**, a hybrid decision-making emotional support agent that integrates symbolic reasoning with deep learning. This model combines rational inference with emotional empathy, enabling the agent to generate supportive responses that balance logic and emotion. Specifically, we introduce **Sym-Mind**, a differentiable logic-based reasoning framework for emotional support strategy selection, which unifies interpretability with stable performance. Experimental results on public datasets demonstrate that our approach consistently outperforms multiple competitive baselines in both automatic and human evaluations, validating its effectiveness.

CCS Concepts

• **Theory of computation** → **Semantics and reasoning**; • **Computing methodologies** → **Artificial intelligence**.

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Keywords

Natural Language Processing; Emotion Support Conversation; Large Language Model; Symbolic Reasoning

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Resource Availability:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18361130>.

1 Introduction

Emotional response generation focuses on understanding a user's emotional states and experiences in dialogue, and generating empathetic and helpful responses[21]. In recent years, it has demonstrated promising results across multiple domains[7, 23, 55], including automated psychological counseling[33, 43], role-playing patients[35], and developing conversational agents[14, 31]. However, the complexity and sensitivity of emotional conversations require not only rule-adherent responses but also deep emotional understanding to provide timely emotional feedback[9, 22], especially when users experience strong emotional fluctuations. Drawing on Dual-Process Theory[26], System 1 operates rapidly, intuitively, and emotionally, whereas System 2 is slower, deliberative, and logic-driven. Emotional support dialog systems should therefore strive to balance emotional responsiveness with rational reasoning[58].

Beyond response quality, a reliable emotional support dialogue system should not only generate high-quality responses but also ensure interpretability[53, 54]. In particular, we do not know whether the reasoning processes of these systems truly conform to established support theories[54], which makes it difficult to determine

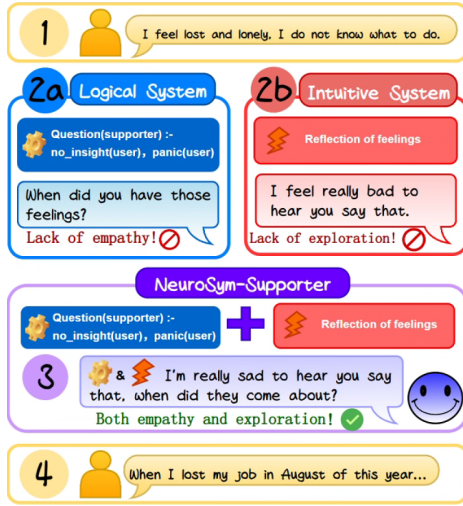


Figure 1: Logical systems tend to directly address problems, while intuitive systems exhibit greater empathy. Ours combines the strengths of both thinking modes to generate balanced responses.

whether systems claiming to be interpretable are actually reasoning rationally. The black-box nature limits the further development and practical application of such support systems[54]. Consequently, interpretable emotional support dialogue systems have attracted increasing attention[54]. For example, Tu et al. [48] introduces a CoT-based fine-tuning approach that incorporates commonsense knowledge graphs[5], while Chen et al. [8] uses a cause-aware CoT to guide the model in reflecting on the reasons behind the user’s emotions. Zhang et al. [53] simulates the process of human emotional support generation—including emotion identification, understanding, and regulation—using reasoning chains to produce explainable responses. Chen et al. [6] constructed a mental health counseling dataset with multitask labels and explanations of the dialogue process. However, current CoT-based methods are constrained by the inherent instability of LLMs[8], as their reasoning processes heavily rely on generative capabilities[53], resulting in unstable inference and limited interpretability.

To address the aforementioned challenges, we propose Neuro-Sym Mind, a hybrid decision-making framework inspired by Dual-Process Theory [26], which integrates symbolic reasoning with deep learning. As illustrated in Figure 1, this framework enables the agent to combine rational and deliberate inference [40] with sensitivity to emotional cues [16], thereby generating supportive responses that balance logic and empathy. Building upon this framework, we develop Neuro-Sym Supporter (NSS), an emotional support agent capable of accurately recognizing users’ mental states and producing high-quality, interpretable emotional responses. Overall, the agent aims to achieve interpretable reasoning and emotionally grounded support through the synergy of rational cognition and affective understanding.

However, effectively integrating symbolic reasoning with neural learning presents several challenges. Traditional neuro-symbolic systems often rely on task-specific networks to output truth values

for logical states [45, 47], but such models lack robust textual understanding and flexibility. Moreover, in complex dialogue scenarios, selecting the optimal rule from a large set of candidates can be hindered by rule interference, where suboptimal rules distort the reasoning process [3, 18]. To overcome these issues, we introduce Sym-Mind, a differentiable logic-based reasoning method for emotional support dialogues. Specifically, we model the support process as symbolic reasoning over seven logical states derived from [22], and introduce a modified sigmoid function (termed “confidence-sigmoid”) for truth-value processing. This modification ensures the differentiability of reasoning operations while reducing interference from secondary rules, thus achieving both interpretability and stable rational inference. Experiments on two empathetic dialogue datasets demonstrate that our approach surpasses mainstream baselines in strategy prediction accuracy and response quality, and the analysis of strategy distributions further verifies the effectiveness and generalizability of our hybrid design.

In summary, our main contributions are as follows:

- (1) We propose Neuro-Sym Supporter, an emotional support agent built on a hybrid decision-making framework that effectively combines rational reasoning and emotional empathy.
- (2) We introduce Sym-Mind, which incorporates a confidence-sigmoid to implement a simple yet effective differentiable logic-based strategy reasoning method, thereby enhancing the interpretability of emotional support dialog systems.
- (3) Extensive experiments on two widely used public datasets demonstrate the effectiveness of our approach and further validate both the effectiveness and necessity of the hybrid design.

2 Related work

Emotional Support Conversation. Emotional support conversation (ESC)[33] is a multi-stage emotional support framework based on Hill and O’Brien [22], which divides the support process into the stages of exploration, insight, and action. It combines multiple support strategies to simulate natural interaction patterns and provide effective emotional support. With the rise of LLM-based technologies[27, 37], LLM-based emotional support systems have gradually become mainstream approaches[10, 23, 32], including strategy-based methods[7, 11, 42], knowledge-augmented models[15, 25, 34], and multi-agent collaborative systems[12, 44]. Recently, some works have attempted to build interpretable emotional support systems[8, 53, 54], but these methods still heavily depend on the performance of LLMs[53]. Building an emotional support system that maintains high performance and interpretability remains a major challenge[53, 54].

Deep Learning. Deep learning leverages neural networks to learn optimal decision-making policies in complex environments[2, 29, 38]. It has been applied to natural language processing tasks including dialogue systems[29], information extraction[24], and text classification[19, 49, 51]. In emotional support dialogues, deep neural networks are often used for strategy selection[28, 50, 58]. However, these works still face interpretability challenges[13].

Neural-symbolic Reasoning. Neural-symbolic reasoning is an artificial intelligence approach that combines the learning capabilities of neural networks with the interpretable reasoning of symbolic systems[1], which has achieved significant progress over the

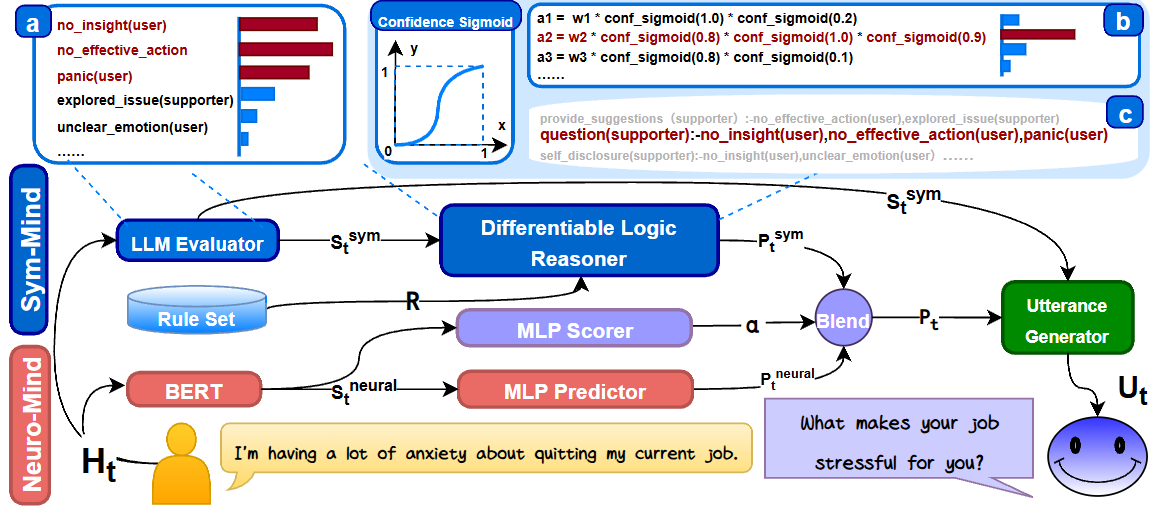


Figure 2: Overview of Neuro-Sym Supporter. Sym-Mind conducts (a) fact extraction, (b) differentiable reasoning, and (c) rule matching to produce the logical strategy distribution. Neuro-Mind encodes the dialogue history through BERT and generates neural preferences using an MLP predictor. Both outputs are fused by a learnable weight into the final strategy distribution, which guides the utterance generator to produce the empathetic response.

past decade[17, 20]. Recent systems have been applied to various tasks, such as visual question answering[36], decision-making in embodied intelligence[57] and video reasoning[10]. Delfosse et al. [13], Shindo et al. [46] represent policy learning as a set of weighted rules learned via policy gradient methods, an approach later extended to interpretable reinforcement learning agents by Shindo et al. [45]. However, building emotional support agents using such methods remains an open research problem[54].

3 Preliminaries

Problem Formulation. We model the dialogue as a sequential decision-making process. At each turn t , given the dialogue history $\mathcal{H}_t = \{u_1, s_1, \dots, u_t\}$, where u_t and s_t denote user and system utterances, the system first selects an optimal strategy a_t and then generates a response U_t conditioned on it. This can be compactly written as $a_t^* = \arg \max_{a_t \in \mathcal{A}} \text{Sim}(a_t, a_t^{\text{gt}})$ and $U_t^* = \arg \max_{U_t} \text{Sim}(U_t, U_t^{\text{gt}} | a_t^*)$, where $(a_t^{\text{gt}}, U_t^{\text{gt}})$ are the ground-truth strategy and response, and $\text{Sim}(\cdot)$ measures similarity to the ground truth.

Differentiable Reasoning. Differentiable reasoning transforms traditional first-order logic into a differentiable framework. In this setting, a set of rules $\mathcal{R}$ with weights $\mathbf{W}$ is mapped to a probability distribution over candidate strategies such that $P_{\text{sym}}(a_t | s_t^{\text{sym}}; \mathcal{R}, \mathbf{W}) = f^{\text{reason}}(s_t^{\text{sym}}; \mathcal{R}, \mathbf{W})$, where s_t^{sym} is the symbolic state vector of the dialogue context, f^{reason} denotes a reasoning function.

4 Method

As illustrated in Figure 2, the Neuro-Sym Supporter integrates rational reasoning and sub-symbolic reactions by combining symbolic and neural policies. Sym-Mind performs differentiable reasoning over symbolic states (§4.2), while Neuro-Mind computes a distribution over actions from sub-symbolic representation (§4.3). These

two distributions are then fused to form the final action distribution (§4.4), which, along with the symbolic states, is fed into an utterance generator to produce the response (§4.5).

4.1 Hybrid State Representations

Agents employ two distinct state representations: (i) object-centric symbolic representation and (ii) BERT-based neural representation. The symbolic representation consists of a list of objects, including the supporter and the seeker, along with their attributes (e.g., emotional state, degree of exploration), enabling logical reasoning over structured information. The neural representation encodes the dialogue history using BERT; the embeddings are fed into a multi-layer perceptron (MLP) predictor for strategy prediction.

Formally, in turn t , the symbolic state is represented as $s_t^{\text{sym}} = \mathbf{z} \in \mathbb{R}^{n \times m}$, where n denotes the number of objects and m the number of extracted properties. Each entry of $\mathbf{z}$ corresponds to a specific property of an object. The symbolic state is obtained via a learned fact perception module, which extracts structured symbolic information from the dialogue context $\mathcal{H}_t$ and produces a probabilistic symbolic state suitable for differentiable reasoning.

Similarly, the neural state s_t^{neural} consists of the embeddings of the recent T utterances, denoted as $s_t^{\text{neural}} \in \mathbb{R}^{T \times d}$, capturing both semantic and contextual information across multiple turns.

4.2 Sym-Mind

4.2.1 Logic Programs for Actions. We define an Action-State Language as a tuple $(\mathcal{P}_A, \mathcal{P}_S, \mathcal{D})$, where $\mathcal{P}_A$ is a set of action predicates (e.g. `provide_suggestions`, `question`), $\mathcal{P}_S$ is a set of state predicates capturing user psychological conditions (e.g. `no_insight`, `need_information`) and $\mathcal{D}$ is a domain-specific set of constants (e.g. `supporter`, `seeker`). Each symbolic rule encodes a condition-action mapping in the form: $X_A : X_S^{(1)}, \dots, X_S^{(n)}$, where X_A is the target

action atom and $X_S^{(i)}$ are the preconditions expressed as symbolic state atoms. For example: $\text{question(supporter)} :- \text{no_insight(user)}, \text{panic(user)}$. This rule reflects the following principle: If the user is panicking and lacks insight into their emotional cause, the supporter should ask a question to facilitate exploration.

To construct the symbolic rule set C , we employ a hybrid method combining data-driven mining from the ESConv dataset and psychological theory from [22] (§B.2).

4.2.2 Differentiable Logic Policies. To enable Sym-Mind to capture the complex semantics of dialogue context, we employ an LLM as the perception function:

$$\mathbf{v}^{(0)} = \text{LLM}(\mathcal{H}_t), \quad (1)$$

where each element $v_i^{(0)} \in [0, 1]$ represents the confidence that the corresponding fact f_i holds. For example, given $\mathcal{G} = \text{panic(user)}, \text{no_insight(user)}$ the confidence vector $\mathbf{v}^{(0)} = [0.8, 0.6]$ indicates the initial belief degrees of these facts.

To map these probabilistic facts to rule activations, we introduce a fixed-size rule set C with M rules and associate it with a learnable weight vector $\mathbf{W} = [w_1, \dots, w_M]^T$, enabling Sym-Mind to acquire policy learning capability via optimization of these weights.

However, directly training the weights using the initial confidence poses two challenges: (i) the LLM may assign zero confidence to certain facts, which would cause crisp logical evaluation to yield zero rule activation and result in vanishing gradients during training; and (ii) different actions may be supported by varying numbers of rules, causing actions with more rules to be favored by aggregated confidence scores, even when a single highly relevant rule for another action provides stronger evidence, thereby introducing noise from weakly matched or irrelevant rules. To address these challenges, we introduce the confidence sigmoid:

$$\sigma_k(x) = \frac{1}{1 + e^{-k(x-0.5)}}, \quad (2)$$

where k controls the sharpness of the activation. As $k \rightarrow \infty$, σ_k approaches a step function, while finite values of k yield a smooth, differentiable transition. An appropriate choice of k mitigates vanishing gradients during training while suppressing the influence of partially satisfied rules. The activation strength of a rule r_j is computed as the differentiable conjunction over its premises:

$$\mu(r_j) = \prod_{f_i \in \text{premise}(r_j)} \sigma_k(v_i), \quad (3)$$

where $\text{premise}(r_j)$ denotes the set of predicate symbols in the rule antecedent. For the differentiable reasoning function, the score of each candidate action $a \in \mathcal{A}$ is obtained by a weighted sum over all rules whose conclusions correspond to that action:

$$s(a) = \sum_{r_j \in \mathcal{R}_a} w_j \cdot \mu(r_j). \quad (4)$$

Finally, we apply a softmax function to normalize the action scores and obtain the symbolic policy distribution:

$$P_{\text{sym}}(a_t | s_t^{\text{sym}}; \mathbf{W}) = \text{softmax}(s(a_t)). \quad (5)$$

4.3 Neuro-Mind

Neuro-Mind employs a neural strategy predictor based on a MLP, chosen as a simple and efficient way to cast strategy prediction into policy learning:

$$z_t = f_a(s_t^{\text{neural}}; \theta_{\text{neural}}), \quad (6)$$

where f_a denotes the classifier, s_t^{neural} denotes the neural representation of the dialogue history. The resulting logits are transformed into an action distribution:

$$P_{\text{neural}}(a_t | s_t^{\text{neural}}; \theta_{\text{neural}}) = \text{softmax}(z_t). \quad (7)$$

4.4 Neuro-Sym Mind

Neuro-Sym Mind computes the final strategy distribution by performing a convex combination of the output of the two reasoning modules. A gating scalar $\alpha \in [0, 1]$ is dynamically computed from the dialogue history:

$$\alpha = \sigma(f_b(s_t^{\text{neural}}; \theta_\alpha)), \quad (8)$$

where f_b denotes an MLP scorer, s_t^{neural} is the BERT-based neural representation of the dialogue history up to turn t , and the sigmoid function $\sigma(\cdot)$ ensures that $\alpha \in [0, 1]$. The supervision of α is derived from its alignment with the gold strategy labels: when P_{sym} aligns more closely with the true strategy than P_{neural} , α increases, and vice versa. The final policy is a convex combination:

$$P(a_t | s_t) = \alpha \cdot P_{\text{sym}}(a_t | s_t^{\text{sym}}; \mathbf{W}) + (1 - \alpha) \cdot P_{\text{neural}}(a_t | s_t^{\text{neural}}; \theta_{\text{neural}}), \quad (9)$$

where P_{sym} is the Sym-Mind strategy distribution, parameterized by rule weights $\mathbf{W}$; P_{neural} is the Neuro-Mind distribution, parameterized by neural parameters θ_{neural} ; and $\alpha \in [0, 1]$ is a context-sensitive blending coefficient. We optimize the rule weights $\mathbf{W}$, neural parameters θ_{neural} , and gating parameters θ_α using cross-entropy loss:

$$\mathcal{L}(a_t, \hat{a}_t) = - \sum_{k=1}^{|\mathcal{A}|} a_{t,k} \log \hat{a}_{t,k}, \quad (10)$$

where $a_{t,k} \in \{0, 1\}$ represents the one-hot encoded ground truth label and $\hat{a}_{t,k}$ is the predicted strategy probability distribution.

4.5 Utterance Generator

We explore two approaches to implementing the utterance generator: a fine-tuned approach and a prompt-based approach. Intuitively, the former directly learns the nuanced patterns of complex dialogue tasks from the dataset, while the latter leverages the strong capabilities of LLMs and is expected to generalize better across diverse scenarios. In static experiments, the fine-tuned approach is based on Qwen3-14B; we also fine-tune multiple models to analyze the impact of model family and parameter size on ESC performance (§5.4.2). Specifically, the input to the generator is formed by concatenating the dialogue context $\mathcal{H}_t$, the current strategy a_t , and the current symbolic state s_t^{sym} . The generator is trained to minimize the generation loss $\mathcal{L}_G$, defined as the negative log-likelihood of the ground-truth tokens. For the prompt-based approach, we

Table 1: Static evaluation results on the ESConv.

Generation Paradigm	Model	ACC	BL-1	BL-2	BL-4	RG-L	MET	Dist-1	Dist-2	Dist-3
Prompt-based	GPT-5	29.33	15.13	4.85	1.83	14.32	15.63	2.86	18.63	37.30
	GPT-5+CoT	28.21	14.84	4.67	1.78	14.78	15.17	2.98	19.42	39.45
	COOPER _(PT)	-	17.62	5.30	1.85	15.65	16.28	5.34	30.66	56.17
	NSS _(PT)	40.66	18.34	6.20	2.14	16.16	16.82	3.01	19.28	40.41
Fine-tuned	BlenderBot-Joint	28.34	16.93	9.97	1.74	16.39	14.78	4.12	17.72	34.51
	MISC	31.30	18.73	7.31	2.20	17.91	16.67	4.41	19.71	43.71
	KEMI	-	20.94	8.63	2.67	17.48	17.31	2.77	15.26	30.22
	COOPER _(FT)	-	22.81	9.60	3.13	20.24	19.92	4.95	23.50	42.48
	NSS _(FT)	40.66	24.23	10.16	3.47	22.28	21.43	5.32	25.22	47.79

Table 2: Static evaluation results on the EmpatheticDialogues.

Generation Paradigm	Model	BL-1	BL-2	BL-4	RG-L	MET	Dist-1	Dist-2	Dist-3
Prompt-based	GPT-5	12.63	5.09	1.78	13.20	14.41	2.90	20.23	40.40
	GPT-5+CoT	12.40	5.56	1.88	13.78	14.69	3.18	22.42	45.20
	COOPER _(PT)	14.86	5.88	1.81	15.72	16.66	4.72	27.61	48.20
	NSS _(PT)	15.28	6.31	1.98	16.06	17.11	3.24	18.98	39.66
Fine-tuned	CEM	14.35	5.84	2.55	15.88	15.31	2.57	14.98	30.21
	COOPER _(FT)	16.46	7.22	3.06	18.20	17.41	5.02	23.87	41.76
	NSS _(FT)	19.37	7.98	3.45	20.45	21.04	5.21	24.28	45.55

directly use an LLM to generate the next utterance $\mathcal{U}_t$, with the prompt including the dialogue history, strategy information, and the current symbolic state. In the following, we refer to the Neuro-Sym Supporter with the fine-tuned generator as **NSS_(FT)**, and the one with the prompt-based generator as **NSS_(PT)**.

5 Experiments

5.1 Experimental Setup

5.1.1 Datasets. Our experiments are conducted on the ESConv[33] and EmpatheticDialogues[41]. ESConv is an ESC dataset that includes 1,300 conversations, each annotated with the support strategies used by the supporter. We follow the setting in [33] for its data pre-processing and data split. After pre-processing, the training/validation/test sets contain 1,040/130/130 conversations, respectively. EmpatheticDialogues includes 24,850 conversations, each annotated with the emotion label of the speaker. We follow the setting in [41] for its data pre-processing and data split. After pre-processing, the training/validation/test sets contain 19,533/2,770/2,547 conversations, respectively.

5.1.2 Baselines. Our baseline methods include two LLM-based prompting approaches and several state-of-the-art models. Specifically, we evaluate the following prompt-based methods: **GPT-5** generates responses by conditioning on a brief task description and the dialogue history, using a prompt format similar to [56]; **GPT-5+CoT** extends GPT-5 with Chain-of-Thought reasoning [52] to infer the user’s state and select an appropriate strategy before response generation. We also compare against fine-tuned generative models, including: **BLENDERBOT-JOIN** [33], which incorporates a strategy token at the start of generation to improve response relevance; **MISC** [48], which enhances emotion understanding with

commonsense knowledge and uses a mixed strategy for empathetic response generation; **KEMI** [15], which improves proactivity and empathy by retrieving mental health knowledge from a large-scale graph; **COOPER** [12], which coordinates multiple specialized agents to handle different aspects of the dialogue and synthesizes their outputs for coherent guidance; and **CEM** [42], which employs cognitive and affective reasoning over commonsense knowledge to understand both emotion and context, generating more informative and empathetic responses.

5.1.3 Implementation Details. All of our training, including the strategy reasoner and LLMs fine-tuning, was conducted on a single 80GB A100 GPU. All prompt-based modules in our approach, as well as prompt-based baseline models, are implemented using GPT-5. For other baselines, we performed experiments using their publicly released code. When evaluating strategy prediction accuracy, we use the hybrid model to output only the strategy with the highest probability. Experiments are repeated five times, and the results are averaged. However, during text generation, we simultaneously consider the optimal strategies output by both Neuro-Mind and Sym-Mind as a combined strategy. For instance, if one strategy advances the support process (e.g., Question) and the other focuses on emotional comfort (e.g., Self-disclosure), we combine both to generate responses that are both effective and emotionally balanced. The advantages of this approach are demonstrated in the case study (§A). ALL prompts are provided in the appendix (§D).

5.2 Static evaluation

We conduct a static evaluation on the generated utterances by comparing them with the ground-truth ones in the datasets. We use the following automatic metrics: strategy prediction accuracy (**ACC**);

Table 3: Interactive evaluation results (%). The columns of “Win/Lose” indicate the proportion of cases where the former model in that set of comparisons wins/loses. †/‡ denote p -value $< 0.1/0.05$ (statistical significance test).

Compared Models	Coherence		Natural		Identification		Empathy		Suggestion	
	Win	Lose	Win	Lose	Win	Lose	Win	Lose	Win	Lose
NSS _(FT) vs. GPT-5	36.2	33.8	55.3 [‡]	28.2	20.4	15.2	63.2 [‡]	15.9	57.2 [‡]	21.8
NSS _(FT) vs. COOPER _(FT)	38.1 [†]	35.7	36.4 [†]	30.9	19.6	23.2	41.0 [‡]	30.2	33.4	28.9
NSS _(FT) vs. NSS _(PT)	42.8 [‡]	21.1	44.2 [†]	34.6	27.5 [†]	24.5	42.6 [†]	27.8	37.3 [†]	29.6

Table 4: Ablation study on the ESConv dataset.

Model	ACC	BL-1	BL-2	RG-L	MET
w/o S-M	31.02	20.03	9.01	19.66	18.33
w/o N-M&S _L	31.28	20.88	9.31	19.03	18.71
w/o N-M	31.28	22.47	9.59	20.43	19.87
NSS _(FT)	40.66	24.23	10.16	22.28	21.43

BLEU-1/2/4 (**BL-1/2/4**) [39], which measure the n-gram precision; ROUGE-L (**RG-L**) [30], which measures the recall of longest common subsequences; METEOR (**MET**) [4], which further considers stem match or synonymy match; Distinct-1/2/3 (**Dist-1/2/3**), which calculates the ratios of unique n-grams.

5.2.1 Comparison with Baselines. The evaluation results on the ESConv and EmpatheticDialogues datasets are presented in Tables 1 and 2, respectively. For clarity, we classify the compared models into two categories based on their utterance generation paradigms: LLM prompt-based methods and fine-tuned methods. In both categories, the two variants of our framework (NSS_(PT) and NSS_(FT)) outperform the baseline models in overall performance across both datasets, demonstrating the effectiveness of our approach. In particular, our models achieve significant improvements in ACC, indicating more precise strategy prediction, which in turn enhances performance on other metrics.

Within the prompt-based methods, NSS_(PT) performs significantly better on the ESConv dataset in BLEU-1, BLEU-2, BLEU-4, ROUGE-L, and METEOR. This indicates that the generated responses are more consistent with the reference data. We attribute this improvement primarily to increased strategy prediction accuracy. Additionally, by explicitly incorporating the user’s logical state into the prompt, the model is guided to generate responses that better address user needs. NSS_(PT) achieves comparable performance to the baselines in Dist-1, Dist-2, and Dist-3. This may result from the fact that our responses are grounded in predefined strategies and contextual information, where limited strategy cues constrain output diversity. Notably, our model underperforms COOPER_(PT) in Dist-2, likely because COOPER’s focus on multiple dialogue goals introduces richer linguistic variations, thereby enhancing response diversity. In the fine-tuned category, NSS_(FT) achieves the best performance on ESConv, again primarily due to improved strategy prediction accuracy. Moreover, by incorporating the user’s logical state into the input during fine-tuning, the model can better understand user needs and mitigate potential strategy errors.

Our proposed methods maintain superior performance on the EmpatheticDialogues dataset. Among prompt-based approaches,

NSS_(PT) achieves the highest scores in BLEU-1, BLEU-2, BLEU-4, ROUGE-L, and METEOR, while NSS_(FT) achieves the best performance across all metrics. COOPER performs below expectations on EmpatheticDialogues, which we attribute to the shorter and simpler dialogue turns in this dataset, limiting its potential to address complex conversational challenges.

In general, our methods demonstrate strong performance on both datasets, which confirms the generalizability of our approach. Furthermore, fine-tuned methods generally outperform prompt-based methods. However, due to stronger supervision from the training data, this does not imply the absolute superiority of fine-tuning. Therefore, we conducted an interactive evaluation to assess the performance of the model in real-world scenarios (§5.3).

5.2.2 Ablation Study. To investigate the contribution of different modules in our framework, we conduct ablation studies on the ESConv dataset. The complete NSS_(FT) framework is compared against the following variants: (1) w/o S-M: The Sym-Mind module is removed, and Neuro-Mind is used directly as the strategy predictor; (2) w/o N-M: The Neuro-Mind module is removed, and Sym-Mind is used directly as the strategy predictor; (3) w/o N-M&S_L: This variant evaluates the impact of incorporating logical states into the utterance generator. To isolate this factor, we remove the Neuro-Mind module and exclude the symbolic state from the generator’s input during fine-tuning.

As shown in Table 4, NSS_(FT) achieves the best performance across all metrics. Removing any component leads to a performance drop, indicating the indispensability of each module. Specifically, the comparison between w/o S-M and w/o N-M&S demonstrates that a decrease in ACC substantially affects overall system performance. Furthermore, the comparison between w/o N-M&S_L and w/o N-M shows that incorporating the logical state as input during fine-tuning significantly improves model performance. We hypothesize that the logical state helps mitigate the negative impact of incorrect strategy predictions during text generation.

5.3 Interactive Evaluation

We further evaluate the proposed methods in an interactive setting by simulating realistic system-user conversations. Fifteen participants from diverse professional backgrounds engaged in discussions covering a total of 50 topics. Interactions were conducted in a free-form manner, allowing participants to terminate the conversation at their discretion. Following the evaluation protocol in [12], we manually assessed system performance across five dimensions: (1) **Coherence**: the consistency of model responses with the dialogue context; (2) **Naturalness**: the naturalness and human-likeness of

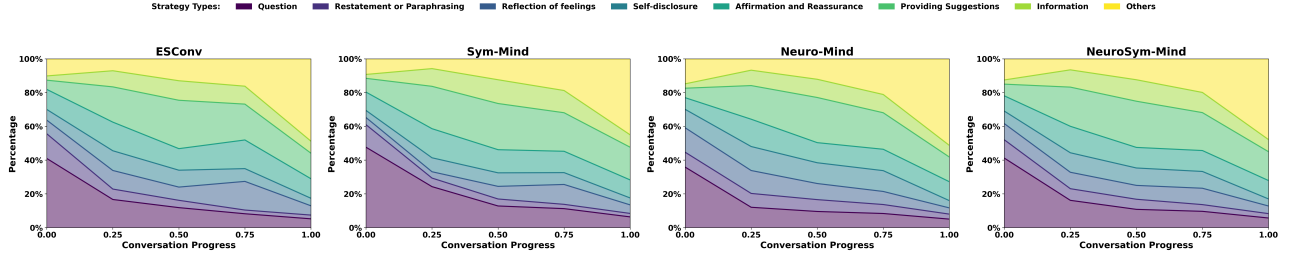


Figure 3: The strategy distribution in the different stage of conversation.

Table 5: Static evaluation results of different models based on different finetuning methods on ESConv dataset.

Generation Paradigm	Model	BL-1	BL-2	BL-4	RG-L	MET	Dist-1	Dist-2	Dist-3
Prompt-based	GPT5+CoT	15.13	5.27	1.83	14.32	14.17	4.86	26.63	53.30
Conventional Finetuned	Llama-3-8B-Instruct	18.12	8.63	3.21	18.58	17.89	4.46	28.27	56.18
	Mistral-7B-Instruct-v0.3	20.02	10.30	5.58	21.06	21.11	5.06	31.21	60.64
	Qwen3-8B	15.64	6.72	2.83	17.32	16.17	3.96	23.21	46.49
	Qwen3-14B	20.24	8.96	3.18	18.28	17.63	5.91	28.06	52.49
	Qwen3-32B	21.62	9.12	3.31	20.16	18.36	6.03	29.45	58.49
NSS Finetuned	Meta-Llama-3-8B-Instruct	22.96	9.37	3.48	19.01	18.28	4.51	28.71	57.42
	Mistral-7B-Instruct-v0.3	23.85	10.74	6.21	22.22	22.09	5.11	30.78	58.24
	Qwen3-8B	18.24	7.43	3.92	19.32	18.87	4.41	26.94	48.87
	Qwen3-14B	24.23	10.16	3.47	22.28	21.43	5.32	25.22	47.79
	Qwen3-32B	24.58	10.64	3.65	23.02	21.86	6.22	33.67	62.84

model responses; (3) **Identification**: the model’s ability to identify the seeker’s underlying issues; (4) **Empathy**: the degree of empathy demonstrated toward the seeker; (5) **Suggestion**: the practical and context-specific relevance of the advice provided. Ten annotators participated, including five graduate students with a linguistics background and five without. We compared two variants of our framework, $NSS_{(PT)}$ and $NSS_{(FT)}$, against two representative baselines from different categories: $COOPER_{(FT)}$ and GPT-5. A direct comparison between the two variants was also conducted to assess relative effectiveness in utterance generation.

As shown in Table 3, $NSS_{(FT)}$ outperforms GPT-5 across all metrics, particularly in naturalness and empathy. GPT-5 tends to generate excessive advice in a didactic tone, often overlooking the supportive aspect of emotional comfort. In contrast, $NSS_{(FT)}$ achieves a balanced performance across all dimensions. Although GPT-5 generates more suggestions, its performance on the Suggestion metric is lower, as its advice is generally too generic and not well adapted to the seeker’s context. $NSS_{(FT)}$ also surpasses the competitive fine-tuned baseline $COOPER_{(FT)}$ in overall performance. Although $COOPER_{(FT)}$ leverages multi-objective optimization to produce more comprehensive suggestions and enhance naturalness, it underperforms $NSS_{(FT)}$ in empathy. We attribute this advantage to two factors: (1) the use of mixed strategies during text generation, combining exploratory strategies with emotional feedback strategies; and (2) the incorporation of the user’s current logical state into the context, allowing the model to account for emotional feedback even when only an exploratory strategy is applied. In the direct comparison between the two variants, $NSS_{(FT)}$ outperforms $NSS_{(PT)}$, particularly in empathy and suggestion, indicating that

fine-tuning enables $NSS_{(FT)}$ to learn richer empathetic expressions and more concrete, context-aware advice strategies. Further analysis shows that the average dialogue length when interacting with $NSS_{(FT)}$ is 9.4 turns, compared to 4.5 turns for GPT-5 and 7.8 turns for $COOPER$, indicating improved user engagement.

5.4 Further Analysis

5.4.1 Strategy Distribution Analysis of Neuro-Sym Mind. To further analyze the advantages of Neuro-Sym Mind over its individual components in strategy selection, we present a stacked percentage line chart in Figure 3, illustrating the distribution of selected strategies on the test set. Sym-Mind exhibits a stronger preference for strategies such as *Question* and *Providing Suggestions*, reflecting its logic-driven approach aimed at directly addressing problems. In contrast, Neuro-Mind favors strategies like *Reflection of Feelings* and *Affirmation and Reassurance*, highlighting the emotional sensitivity and responsiveness of its neural network-based decision-making. Neuro-Sym Mind, which integrates both components, achieves a strategy distribution closer to the ground truth, indicating that combining Sym-Mind and Neuro-Mind effectively balances rational reasoning with emotional feedback. This demonstrates the benefits of integrating symbolic reasoning with deep learning in emotional support dialog systems.

5.4.2 Effect of Model Size and Type on Fine-tuning. To evaluate the generalizability of our method and the impact of parameter scale on the ESC task, we conducted comparative experiments on the ESConv dataset using multiple model architectures, including Llama-3-8B-Instruct, Mistral-7B-Instruct-v0.3, and Qwen-3 (8B/14B/32B).

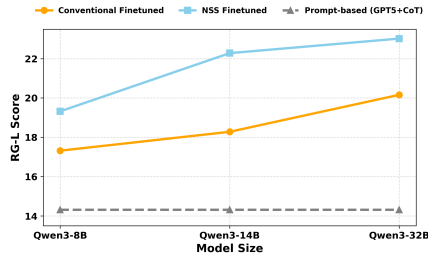


Figure 4: RG-L Scores of Qwen with Different Methods

Table 5 shows that NSS fine-tuning consistently outperforms conventional fine-tuning across all models and metrics, demonstrating strong cross-architecture generalization. Improvements in the Distinct-n (Dist) metrics are relatively modest, suggesting that response diversity depends primarily on the intrinsic capacity of the underlying models. Furthermore, performance gains from scaling the model size exhibit clear diminishing returns, as illustrated in Figure 4. For instance, although Qwen-3 32B achieves substantially higher ROUGE-L scores than Qwen-3 8B, the improvement is limited relative to the four-fold increase in parameters, and gains in diversity metrics are minimal. These results indicate that emotional support tasks rely less on a model’s inherent reasoning capabilities for complex problem-solving. Once the decision-making framework effectively analyzes the user’s psychological state and generates appropriate strategies, the reliance on the model’s intrinsic reasoning and expressive capabilities diminishes. Consequently, integrating smaller LLMs with targeted fine-tuning within the Neuro-Sym Supporter can already achieve excellent performance, offering a more cost-effective solution for ESC tasks.

6 Ethical Considerations

A central ethical concern of this work is to delineate the capability boundaries of automated dialogue systems. These systems should be limited to non-clinical emotional support and should not be involved in high-risk interventions, including professional psychological counseling or clinical decision-making. When these support boundaries are not explicitly defined, users may misinterpret the system as possessing capabilities beyond its intended design. This misperception can lead to over-reliance, misjudgment of system competence, and potential psychological risks.

To mitigate these risks, our framework explicitly constrains the scope of support at the design level. Specifically, the symbolic reasoning module encodes a limited and interpretable set of user state predicates and support strategies, imposing explicit constraints on system behavior and ensuring operation within well-defined and auditable support boundaries. Built on these constraints, NSS employs a learning-based policy selection mechanism to provide appropriate and responsible emotional support while avoiding encroachment into clinical domains.

From an ethical perspective, this design reflects a cautious principle for constructing emotionally supportive agents. Rather than maximizing surface-level expressions of empathy, the system prioritizes transparency, controllability, and bounded support capabilities. We argue that this approach offers a practical and principled foundation for the responsible design and deployment of emotional support dialogue systems.

7 Conclusion, Limitations and Future Work

This paper investigates how to construct a dialogue system that simultaneously achieves interpretability and deep emotional empathy. By integrating symbolic reasoning with deep learning for support strategy prediction, we propose an emotional support agent, Neuro-Sym Supporter. Experimental results demonstrate the effectiveness of our approach, and case studies further corroborate its interpretability in real-world emotional support scenarios.

Despite its effectiveness, NSS exhibits two primary limitations. First, the state predicate space is limited to seven predefined predicates, which constrains the granularity of user state representation. Second, the current rule base lacks sufficient flexibility, limiting the agent’s ability to handle more complex and diverse scenarios.

In future work, we plan to investigate more fine-grained user state modeling, such as incorporating theories of mind, to enhance the representation of users’ internal mental states. In addition, we aim to explore more flexible symbolic reasoning paradigms, including abductive–deductive reasoning, to improve the expressive power and adaptability of the system in complex settings.

8 Acknowledgments

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A Case Study

Table 6 compares the utterances generated by NSS_(FT) with KEMI, COOPER_(FT), and two prompt-based baselines, GPT-5 and GPT-5+CoT, within an ESC. Several observations can be made. First, our strategy selection is interpretable: Sym-Mind selects ‘Question’ as strategy, and the rule indicates that the system identifies the seeker

Table 6: Case study on the ESConv dataset.

Dialogue History	
Seeker	It really, really is. Things just pile up, you know?
Supporter	Totally! It's hard to plan for success when life seems a bit uncertain.
Seeker	It's a combination of things. I have no idea what I'm doing most days, just kind of going through the motions. Kind of like Groundhog Day, repeating the same day over and over with slightly different variations. Trying to get things done and taken care of, thinking I'm doing my best and feeling like it's never good enough. Feeling like I'm failing all the time.
Comparison	
GPT-5	That sounds incredibly exhausting and disheartening. Just making it through each day takes strength, and the fact that you're still pushing forward really says a lot.
GPT-5+CoT	It sounds exhausting to keep pushing through each day and still feel like it's not enough. You're trying so hard, and it makes sense you'd feel this way. Would you like to share more or just have me sit with you?
KEMI	I get how you feel, maybe you could try picking up some hobbies to help you out of it.
COOPER_(FT)	I hear how hard you're trying, even if it doesn't always feel like enough. It must be really discouraging to feel trapped in the same cycle. What gives you even a small sense of relief or comfort during these times?
Logical Strategy	Question (rule: Question(supporter):- no insight(user), no effective action(user), need information(user))
Neural Strategy	Self-disclosure
Response	Sometimes I feel the same way, too caught up in the to-do list, trying to determine what is important and what isn't. Do you ever find yourself trying to find motivation in a mistake you made in order to succeed?
Ground Truth	Ah yes, the cog in the machine sort of vibe. Boy do I feel that. Well, the good thing is you're not alone. The bad thing is societal pressure demands that we be productive or we think we're wasting time. When's the last time you took a day to enjoy something you like to do?

as being in a confused state, lacking insight into the current issue, showing no effective action, and requiring further information. Since user behaviors have not been thoroughly explored, the model appropriately employs a question to gather more detailed information. Second, our model demonstrates empathy: the self-disclosure strategy selected by Neuro-Mind serves as an empathetic response to the emotional confusion of the seeker. In contrast, KEMI leverages its rich knowledge to suggest that the user “pick up some hobbies.” Although informative, this response lacks strong empathetic engagement and appears overly rational. However, GPT-5 has a clear issue of repetitiveness, typically briefly acknowledging the user’s problem followed by a series of generic encouragements, which we argue are not truly effective. GPT-5+CoT and COOPER_(FT) perform better, showing empathy and further exploration of the problem; however, the level of empathy still falls short compared to the self-disclosure strategy adopted by our model. In general, the response from NSS_(FT) not only further explores the situation of the seeker, but also provides appropriate comfort for the sensitive feelings, with a linguistic style closely similar to the ground truth.

B Rules Extraction

B.1 Data-driven Rules Extraction

We propose a data-driven approach to extract commonly used support rules from the ESConv dataset. Specifically, an LLM is used to compute confidence scores for seven logical states for each dialogue turn. Let $\mathcal{S} = \{s_1, s_2, \dots, s_7\}$ denote the set of symbolic states, where each $s_i \in \mathcal{S}$ represents a distinct State Predicates. Given the dialogue history $\mathcal{H}_t$ at turn t , the LLM outputs an initial state

vector $\mathbf{v}^{(0)}$ as described in Eq. 1. The set of active logical states is then defined as:

$$\mathcal{S}_{\text{active}}^{(t)} = \{s_i \in \mathcal{S} \mid v_i^{(0)} \geq \beta\}, \quad (11)$$

where $\beta \in [0, 1]$ is a predefined threshold. These active states are subsequently used to construct data-driven support rules. For each dialogue turn, we associate the active logical states with the multi-label ground-truth strategies a^t . Each rule is represented as $a_j :- s_i$, for every strategy $a_j \in a^t$ and every active state $s_i \in \mathcal{S}_{\text{active}}^{(t)}$. This forms the preliminary rule set:

$$\mathcal{R} = \{(s_i, a_j) \mid a_j \in a^t, s_i \in \mathcal{S}_{\text{active}}^{(t)}\}. \quad (12)$$

Finally, the extracted rules are refined using a frequency-based filtering mechanism with a threshold of $\tau \geq 22$. With $\beta = 0.5$, this procedure yields a total of 97 rules.

B.2 Theory-driven Rules Extraction

We further refine the data-mined rule set based on Hill and O’Brien [22] by removing rules that are inconsistent with the principles of supportive intervention, resulting in 63 representative rules. We summarize the theoretical conditions for applying each of the seven strategies and present them in Table 9.

C Definitions of Predicates

We follow Liu et al. [33] using eight representative strategies as our action predicates and construct seven representative state predicates based on Hill and O’Brien [22]. The predicates and their definitions are presented in Table 7 and Table 8, respectively.

Table 7: State Predicates and Their Definitions

State Predicates	Definition
no_insight(user)	The user lacks insight into the emotional and situational factors contributing to their current state.
no_effective_action(user)	The user has not identified or taken any effective actions to address their issues.
need_information(user)	The user requires additional information or clarification regarding their situation or available resources.
need_trauma_processing(user)	The user may benefit from processing past traumatic experiences.
unclear_emotion(user)	The user’s emotional state is ambiguous or not clearly expressed.
panic(user)	The user is experiencing panic or acute anxiety.
explored_issue(supporter)	The supporter has already explored the core issue with the user.

Table 8: Action Predicates and Their Definitions

Strategy	Condition
question	Used when clarification of the client’s thoughts, feelings, or behaviors is needed. For example, “What were you thinking when you said those words?” or “Tell me more about your thoughts on that matter.”
providing suggestions	Used when the client has already explored their issues and there is sufficient supportive space. Direct guidance involves the helper offering advice or recommendations to the client. For example, “Next time you have such a nightmare, you might try waking yourself up and imagining a new ending: getting angry at the intruder and driving him out of the house.”
affirmation and reassurance	Used to identify and reinforce emotions, encourage catharsis, clarify, instill hope, and promote self-control. Applied when a good therapeutic alliance exists between the helper and the client. For example, “I think you’re doing really well,” or “I believe you will definitely achieve these things.”
self-disclosure	The helper shares their own feelings or emotions in similar situations to identify and reinforce emotions, encourage catharsis, clarify, strengthen feelings, instill hope, and promote self-control. For example, “When I broke up with my boyfriend, I was very sad.”
reflection of feelings	Repeating or rephrasing the client’s statements, including explicitly identifying emotions. This is usually inferred based on the client’s nonverbal behavior, context, or content. Applicable when the client is avoiding feelings (e.g., showing a clear emotion but trying to interrupt), or when emotional unawareness leads to behavioral dysregulation. For example, “You’re very angry that your husband is not at home.”
information	Providing information, opinions, facts, resources, answers, or perspectives. Used when it is confirmed that the client needs information. For example, “Both the employment center and the counseling center can provide career-related information. It might take about 2 hours to complete this test.”
restatement or paraphrasing	Restating or rephrasing what the client has said, focusing on the meaning they are trying to convey, and selecting the most important part of the narrative. For example, “You want to become an effective helper.”

D Prompt

The prompt templates for **GPT-5**, **GPT-5+CoT**, and **NSS_(PT)** are provided in Tables 10, 11, and 12, respectively.

Table 9: Theoretical conditions for applying each of the seven strategies

Action Predicates	Definition
question(user)	Using questions to guide the conversation, helping the user explore their feelings and circumstances more deeply. This encourages thinking and articulation of thoughts and experiences.
providing_suggestions(user)	Offering practical advice and solutions based on the user's specific situation, aiding them in finding ways to address their issues. Effective after understanding the user's problem thoroughly.
affirmation_and_reassurance(user)	Acknowledging the user's efforts and progress with positive reinforcement and providing comfort and support to boost confidence and motivation for taking effective actions.
self-disclosure(user)	Sharing relevant personal experiences or feelings by the supporter to build trust and empathy, reducing the user's sense of isolation and encouraging openness.
reflection_of_feelings(user)	Explicitly recognizing and responding to the user's emotional state, making the user feel understood and accepted, which aids in acknowledging and accepting one's emotions.
information(user)	Providing users with relevant facts, data, or knowledge to better understand the nature of their problems and possible solutions, supporting informed decision-making.
restatement_or_paraphrasing(user)	Rephrasing the user's words to confirm understanding accurately and show that they are being listened to and understood, which helps clarify information and reduce misunderstandings.
others(user)	Other strategies.

Table 10: Prompts for GPT-5

Function	Prompt
Strategy Selection	<Dialogue History> <Definition and Examples of Strategies> Based on the dialogue history and user's emotional state, which support strategy should be used? Choose from: <List of Strategies>
Text Generation	<Dialogue History> <Definition and Examples of Chosen Strategies> Given the following dialogue history and selected strategy, generate a natural and empathetic response that aligns with the chosen strategy. Ensure the response is supportive and relevant to the user's current emotional state.

Table 11: Prompts for GPT-5+CoT

Function	Prompt
Strategy Selection	<Dialogue History> <Definition and Examples of Strategies> Analyze the user's emotional state based on the dialogue history. Perform step-by-step reasoning to determine the most appropriate support strategy. Explain your reasoning before making a decision.
Text Generation	<Dialogue History> <Definition and Examples of Chosen Strategies> After completing your reasoning, generate a response that implements the chosen strategy. The response should reflect your analysis and be both empathetic and constructive.

Table 12: Prompts for ours Generator

Function	Prompt
Text Generation	<Dialogue History> <Definition and Examples of Chosen Strategies> <Current Logical State and their definition> Given the dialogue history, the selected strategy, and the current logical state, generate a natural and empathetic response that aligns with the strategy in content and resonates emotionally with the user's current state.